

TOPIC: DIABETES PREDICTION USING MACHINE LEARNING

ABSTRACT

Diabetes is a common chronic disease that can lead to serious health complications if it is not detected and managed at an early stage. This project, “Diabetes Prediction Using Machine Learning,” aims to develop a machine learning-based system for predicting whether a person is likely to have diabetes based on relevant medical and demographic parameters. The system uses a dataset containing features such as pregnancy history, glucose level, blood pressure, skin thickness, insulin level, BMI, diabetes pedigree function, and age. The dataset is preprocessed to handle missing or inconsistent values and is then divided into training and testing sets. Machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) can be trained and evaluated to identify the most suitable model for prediction. Performance is measured using metrics such as accuracy, precision, recall, F1-score, and confusion matrix. The proposed system provides a quick and cost-effective preliminary prediction of diabetes risk. It can serve as a decision-support tool for early screening, while the final diagnosis should always be made by qualified healthcare professionals. The project demonstrates how machine learning can be applied to healthcare data to support early detection and improve preventive healthcare.